PatchCvT: Memory-Conditioned Activation-Aware Vision Transformer

1 Introduction

PatchCvT is a memory-conditioned transformer encoder designed to preserve semantically meaningful degraded representations through contextual reasoning.

Unlike traditional anomaly detection methods, PatchCvT does not treat deviation as automatic rejection. Instead, the architecture estimates whether degraded representations remain semantically useful for downstream perception.

2 Input Representation

The input image is defined as:

$$x \in \mathbb{R}^{H \times W \times 3} \tag{1}$$

where the image may be represented in RGB or YUV space.

3 CNN Feature Extraction

A convolutional backbone extracts latent feature maps:

$$F = f_\theta(x) \tag{2}$$

where:

$$F \in \mathbb{R}^{H' \times W' \times C} \tag{3}$$

Patch embeddings are extracted from the feature maps:

$$Q = \{q_j\}_{j=1}^{N} \tag{4}$$

where:

$$q_j \in \mathbb{R}^C \tag{5}$$

4 Memory Bank Construction

A memory bank containing representative semantic priors is defined as:

$$\mathcal{M} = \{m_k\}_{k=1}^{K} \tag{6}$$

where:

$$m_k \in \mathbb{R}^C \tag{7}$$

5 Memory Retrieval

Distances between patch embeddings and memory vectors are computed as:

$$d_{jk} = ||q_j - m_k||^2 \tag{8}$$

Soft retrieval weights are defined as:

$$\alpha_{jk} = \frac{\exp(-d_{jk}/\tau)}{\sum_{l=1}^{K} \exp(-d_{jl}/\tau)} \tag{9}$$

where:

- τ controls retrieval temperature

The retrieved semantic prior becomes:

$$\tilde{q}_j = \sum_{k=1}^{K} \alpha_{jk} m_k \tag{10}$$

6 Residual Degradation Signal

Deviation from retrieved semantic normality is defined as:

$$r_j = q_j - \tilde{q}_j \tag{11}$$

The residual encodes degradation and semantic inconsistency relative to the learned memory priors.

7 Activation Energy Extraction

Activation-aware semantic energy is extracted directly from latent CNN feature maps:

$$a_j = ||F_j||_2 \tag{12}$$

where:

- a_j represents latent semantic activation intensity
- stronger activations indicate semantically relevant structures

8 Semantic Token Construction

PatchCvT tokens are constructed by combining:

- observed patch embeddings
- retrieved memory priors
- residual degradation signals
- activation energy

The final token representation becomes:

$$z_j = [q_j \;||\; \tilde{q}_j \;||\; r_j \;||\; a_j] \tag{13}$$

where $||$ denotes feature concatenation.

9 Contextual Transformer Encoding

The semantic tokens are processed through a transformer encoder:

$$z'_j = Transformer(z_j) \tag{14}$$

The transformer performs contextual semantic reasoning over:

- local appearance
- retrieved semantic priors
- degradation residuals
- activation-aware saliency

10 Semantic Robustness Estimation

PatchCvT estimates semantic robustness confidence:

$$R_j = g_\phi(z'_j) \tag{15}$$

where:

- R_j estimates semantic consistency under degradation

11 Semantic Preservation Gating

Instead of suppressing all anomalous embeddings, PatchCvT estimates whether degraded representations remain semantically useful.

The semantic preservation gate is defined as:

$$S_j = \sigma(W_r R_j - W_e ||r_j||) \tag{16}$$

where:

- R_j is semantic robustness confidence
- $||r_j||$ is degradation magnitude
- W_r and W_e are learnable balancing parameters

The final robustness-conditioned representation becomes:

$$\hat{z}_j = S_j z'_j \tag{17}$$

12 Interpretation

PatchCvT preserves degraded but semantically meaningful representations by conditioning transformer reasoning on:

- retrieved memory priors
- degradation residuals
- activation-aware semantic energy
- contextual semantic consistency

Unlike classical anomaly detection architectures, PatchCvT does not assume that deviation implies irrelevance.

Instead, the architecture performs contextual semantic reevaluation before suppression.